

TABLEKEEPER

Double-booking is physically impossible.

A restaurant booking engine where the **database itself** refuses overlapping reservations — no race, retry storm, or buggy deploy can beat it.

1 × 201 CREATED

49 × 409 SLOT_TAKEN

/kill-demo — 50 concurrent requests, one table. Exactly one winner, every time.

THE WHOLE GUARANTEE

```
EXCLUDE USING gist (  
  table_id      WITH =,  
  tstzrange(starts_at, ends_at, '[')])  
WITH && ) WHERE (status IN  
('held', 'confirmed'))
```

'D' half-open ranges — a booking ending 19:00 and one starting 19:00 don't collide. Back-to-back seatings work like a real restaurant.

DARK FACTORY · HACKATHON BUILD · SEPT 2026

TABLEKEEPER

Double-booking is **physically impossible.**

a restaurant booking engine enforced by a Postgres exclusion constraint

Tablekeeper

Kill demoMy bookingsAPIHealthAudit log

[← home](#)

The Kill Demo

One table. N racers. Same instant. **The Chef's Counter** seats exactly one party — the database refuses every other booking with a clean 409, no matter how the requests interleave.

Each racer is a real HTTP request with its own Idempotency-Key — exactly what two browsers, or fifty, would send.

DATE (ARENA LOCAL: AMERICA/NEW_YORK)

SAT 09-26SUN 09-27MON 09-28TUE 09-29WED 09-30THU 10-01FRI 10-02SAT 10-03SUN 10-04MON 10-05

SLOT & RACERS

target: 6:45 PM2 racers8 racers12 racers24 racers50 racers▶ Fire 50 concurrent bookers

1
201 CREATED

49
409 REJECTED

0
OTHER

VERDICT: exactly one booking survived — 1x201, 49x409, 0xanything-else. 50 racers, one table, zero double-bookings. The constraint is doing the deciding.

Winner: 1c06a8a6-ba61-443f-b4c1-eaf60a3abf7b (held, TTL running)

Confirm the winnerCancel winner & re-arm arena

#	RACER	HTTP	OUTCOME
1	Racer-01	201	hold 1c06a8a6-ba61-443f-b4c1-eaf60a3abf7b · code TK-RPLMRUFM · Counter
2	Racer-02	409	SLOT_TAKEN — overlap rejected

THE KILL DEMO · 50 RACERS · ONE TABLE

Fifty racers. One survivor.

Every racer is a **real HTTP request** with its own Idempotency-Key — exactly what two browsers, or fifty, would send. The verdict settles instantly: one booking created, forty-nine honest rejections answered by the database.

1 × 201 CREATED

49 × 409 SLOT_TAKEN

THE WHOLE GUARANTEE

One constraint. No application logic.

Overlapping bookings on the same table cannot be committed — the storage engine itself refuses the transaction.

```
EXCLUDE USING gist ( table_id WITH =, tstzrange(starts_at, ends_at, '[') ) WITH  
&& ) WHERE (status IN ('held', 'confirmed'))
```

`'['` **half-open** → a booking ending **19:00** and one starting **19:00** don't overlap — back-to-back seatings work like a real restaurant. As strict as reality, not stricter.

AS STRICT AS REALITY — NOT STRICTER

Back-to-back seatings work. Overlaps don't.



▲ 18:30 — A ends, B starts: adjacent, legal. ■ 19:30-20:00 — C intrudes on B: the database refuses the commit.

The half-open '['**]**' range: a booking ending **19:00** and one starting **19:00** don't overlap — turn a table in zero minutes, exactly like a real restaurant floor.

Tablekeeper

Kill demoMy bookingsAPIHealthAudit log

← all restaurants

The Chef's Counter

Omakase · 1 Beacon St, Boston

All times shown in restaurant-local time (America/New_York) · tables: Counter (8)

1 · PICK A DATE (RESTAURANT-LOCAL)

SAT09/26

SUN09/27

MON09/28

TUE09/29

WED09/30

THU10/01

FRI10/02

SAT10/03

SUN10/04

MON10/05

TUE10/06

WED10/07

THU10/08

FRI10/09

2 · PARTY SIZE

1 guest

2 guests

3 guests

4 guests

5 guests

6 guests

3 · PICK A TIME (15-MIN GRID · [START, END] SEMANTICS)

11:30 AM
past

11:45 AM
past

12:00 PM
past

12:15 PM
past

12:30 PM
past

12:45 PM
past

1:00 PM
past

5:30 PM
past

5:45 PM
past

6:00 PM
past

6:15 PM
past

6:30 PM
past

6:45 PM
1 free

7:00 PM
1 free

7:15 PM
1 free

7:30 PM
1 free

7:45 PM
1 free

8:00 PM
1 free

8:15 PM
1 free

8:30 PM
1 free

8:45 PM
1 free

9:00 PM
1 free

9:15 PM
1 free

9:30 PM
1 free

Table held. Counter for 2 · 6:45 PM · code TK-JDEPLJFU

Expires in 9:55 — confirm before the TTL, or the sweep returns the table.

Confirm booking

Test idempotent replay

Release

N

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THE NORMAL FLOW FEELS NORMAL

Hold — a ten-minute lease.

Pick a date, a party size, a slot. The table is **held with a live TTL countdown**; walk away and the sweep quietly returns it to the floor. No orphan holds, no ghost bookings.

201 HELD

TTL 10:00

/restaurants/chefs-counter · hold

sweep re-arms the table on expiry

Tablekeeper

Kill demoMy bookingsAPIHealthAudit log

[← all restaurants](#)

The Chef's Counter

Omakase · 1 Beacon St, Boston
All times shown in restaurant-local time (America/New_York) · tables: Counter (8)

1 · PICK A DATE (RESTAURANT-LOCAL)

SAT
09/26

SUN
09/27

MON
09/28

TUE
09/29

WED
09/30

THU
10/01

FRI
10/02

SAT
10/03

SUN
10/04

MON
10/05

TUE
10/06

WED
10/07

THU
10/08

FRI
10/09

2 · PARTY SIZE

1 guest

2 guests

3 guests

4 guests

5 guests

6 guests

3 · PICK A TIME (15-MIN GRID · [START, END] SEMANTICS)

11:30 AM
past

11:45 AM
past

12:00 PM
past

12:15 PM
past

12:30 PM
past

12:45 PM
past

1:00 PM
past

5:30 PM
past

5:45 PM
past

6:00 PM
past

6:15 PM
past

6:30 PM
past

6:45 PM
full

7:00 PM
full

7:15 PM
full

7:30 PM
full

7:45 PM
full

8:00 PM
full

8:15 PM
1 free

8:30 PM
1 free

8:45 PM
1 free

9:00 PM
1 free

9:15 PM
1 free

9:30 PM
1 free

Confirmed ✓ — Counter for 2 at 6:45 PM (2026-09-26). Confirmation code: TKJDEPLJFU. Lookup anytime at [/lookup](#).

Cancel this booking

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N

CONFIRM → CODE

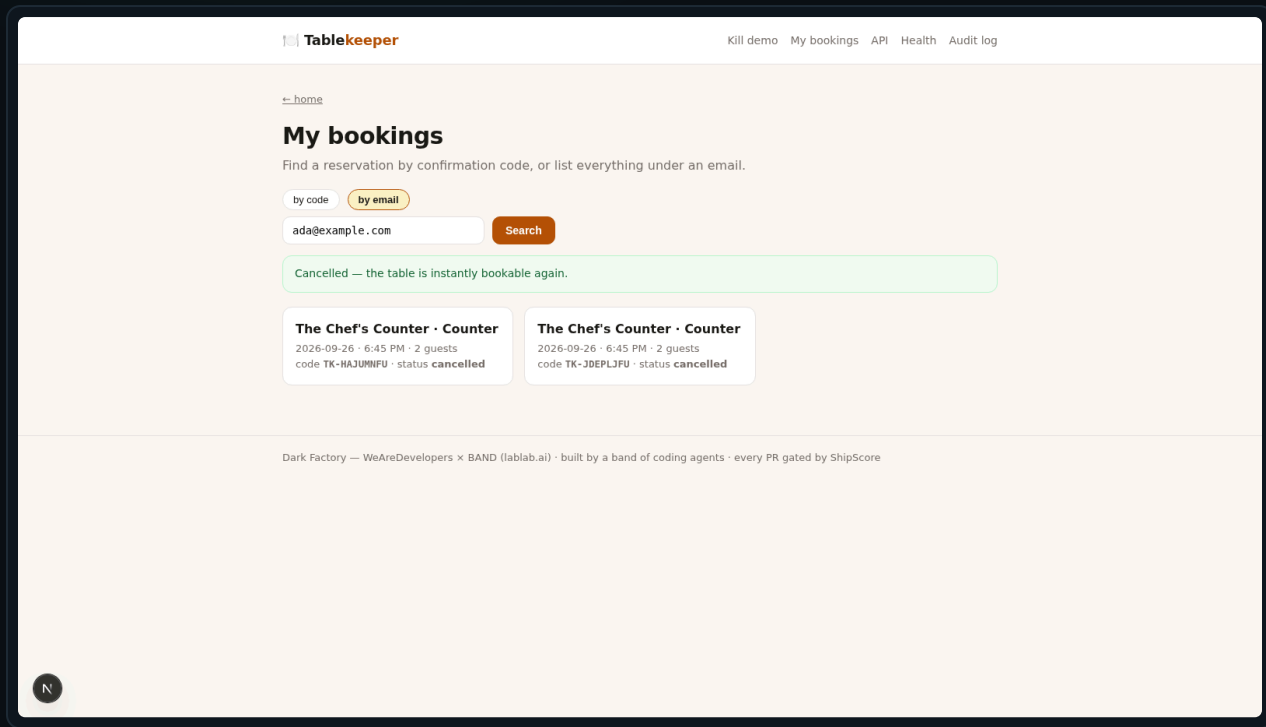
Confirmed. Code in hand.

Confirm converts the hold into a confirmed reservation with a human-readable code.

Replay the same request with the same key — you get the same reservation back. Retries are safe by design.

200 OK

idempotent replay



SELF-SERVICE · AUDITED

Find it. Cancel it. Slot freed.

Guests look up bookings by email and cancel in one click — the slot is **instantly bookable again**. Every transition lands in the audit log with actor, reason and timestamp.

200 CANCELLED

slot re-armed

THE FACTORY CHECKS ITSELF

Every push runs the gauntlet.

28/28

Adversarial suite · T1-T11

50-way race storms, hold expiry, cancel-vs-confirm, DST boundaries, idempotent replay, soak — green on embedded **PGlite** and **postgres:16**.

9/9

Live red team · zero 5xx

Nine probes attack the real HTTP surface — **RED TEAM FAILED TO BREAK IT**. Collisions answered 409, replay 201, expiry 410 — all from the constraint.

100 / A

ShipScore gate

Design · Ship · Run · Secure · Test — scored on every push, merges blocked below threshold. **Day-one honest fail** until the bug it caught was fixed.

Infrastructure, not vibes — **a band of coding agents built the product, and the factory gates its own work.**

RED TEAM FAILED TO BREAK IT.

RT1 · 50-way race, distinct keys

1×201 + 49×409 SLOT_TAKEN — every run

RT2 · 12-way stampede, same key

12×201 → one reservation id

RT3 · same key, different payload

422 DUPLICATE_IDEMPOTENCY_PAYLOAD

RT4a · double confirm

200 + 200 — identical confirm code

RT4b · confirm after terminal cancel

409 INVALID_TRANSITION

RT5 · confirm-vs-cancel race

one coherent terminal state, no 5xx

RT6 · adjacency vs overlap

end==start 201/201 · 45-min overlap 409

RT7 · fuzz battery, 10 invalid clients

all 4xx — zero 5xx

RT8 · health + audit feed

constraintPresent=true, events logged

Zero 5xx across all nine probes — **every verdict comes from the constraint, not from hope.**

Tablekeeper.

Double-booking is impossible.

github.com/icohangar-ops/tablekeeper

Next.js 16 · TypeScript · PostgreSQL + PGlite · Vitest · GitHub Actions · Vercel

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